

SIH26082 — Requirement & Readiness Checklist

Problem: Coupled high-resolution AQI forecasting system for Delhi NCR with a 72-hour outlook.

Use this as the team's PS-compliance checklist before locking architecture and demo scope.

A. PS-DEFINING REQUIREMENTS

Check	Requirement	What to verify
■	Delhi NCR focus	Forecast domain is specifically Delhi NCR.
■	72-hour forecast	Produce AQI forecasts for the next 72 hours.
■	High resolution	Produce spatially detailed forecasts, not one Delhi-wide value.
■	Coupled framework	Use WRF-Chem or a similar open-source coupled weather-chemistry framework.
■	Two-way feedback	Meteorology affects pollution and pollution affects local meteorological conditions.
■	Meteorology	Represent temperature, wind and planetary boundary layer (PBL) height.
■	Chemistry	Represent PM2.5, PM10, O3 and NOx.
■	Atmospheric inversion	Track inversion strength and its pollution-trapping effect.
■	Stubble-burning plume	Predict dispersion of regional stubble-burning pollution under prevailing weather.
■	Real-time dashboard	Provide a user-friendly dashboard with high-resolution forecasts.
■	Actionable output	Explain forecast conditions, not just display AQI numbers.

B. SCIENTIFIC / MODELLING CHECK

Check	Requirement	What to verify
■	Weather to pollution	Wind, temperature and PBL affect transport and dispersion.
■	Pollution to weather	Aerosol/pollution feedback is represented in the coupled framework.
■	Inversion physics	System identifies/quantifies inversion conditions and connects them to pollutant trapping.
■	External pollution	Regional stubble-burning emissions enter the modelling workflow.
■	Plume movement	System can show or quantify plume transport into/through Delhi NCR.
■	72-hour run	The actual coupled workflow generates the required horizon.
■	Spatial output	Model output can be mapped over the Delhi NCR domain.

C. DATA & PIPELINE CHECK

Check	Requirement	What to verify
■	Meteorological inputs	Required weather/atmospheric inputs are available and time-aligned.
■	Pollution inputs	PM2.5, PM10, O3 and NOx observations or model inputs are available.
■	Emission inputs	A defensible source of regional/stubble-burning emissions is identified.
■	Time synchronization	Weather, chemistry and emissions data use compatible timestamps.
■	Spatial alignment	Inputs can be mapped to the model grid.
■	Data ingestion	A repeatable pipeline can fetch, prepare and update inputs.
■	Failure handling	Missing/stale data cannot silently create misleading forecasts.
■	Storage	Forecast runs and observations can be stored for validation.

SIH26082 — Implementation & Demo Checklist

D. AQI & FORECAST OUTPUT

Check	Requirement	What to verify
■	AQI generation	Pollutant concentrations are converted into the intended AQI representation.
■	Pollutant breakdown	Dashboard shows which pollutant drives the forecast.
■	72-hour timeline	User can inspect AQI across the full forecast horizon.
■	Spatial hotspots	High-pollution areas are identifiable on the Delhi NCR map.
■	Forecast drivers	Major drivers such as inversion, low wind or incoming plume are explained.
■	Inversion indicator	Dashboard explicitly displays inversion strength/status.
■	Plume visualization	Dashboard shows predicted stubble-burning plume movement/impact.
■	Forecast updates	Dashboard reflects updated model runs rather than static data.

E. VALIDATION — CRITICAL

Check	Requirement	What to verify
■	Historical backtesting	Test historical periods against observations.
■	Forecast error	Calculate appropriate forecast-error metrics.
■	Spatial validation	Evaluate spatial performance where data permits.
■	72-hour degradation	Measure accuracy from near-term through 72 hours.
■	Pollution episodes	Test high-pollution, inversion and stubble-burning periods.
■	Baseline comparison	Compare against a simpler baseline to show the value of coupling.
■	Reproducibility	A forecast can be repeated from recorded inputs/configuration.
■	Limitations	Team can state uncertainty and known failure cases.

F. DASHBOARD CHECK

Check	Requirement	What to verify
■	Delhi NCR map	Primary spatial visualization is clear.
■	AQI view	Current and forecast AQI are immediately visible.
■	Time control	User can move through the 72-hour forecast.
■	Pollutant layers	PM2.5, PM10, O3 and NOx information is accessible.
■	Wind layer	Wind direction/speed is visible where relevant.
■	PBL/inversion	Atmospheric stability information is understandable.
■	Plume layer	Stubble-burning plume trajectory/impact is clear.
■	Alerts	Potential deterioration can be surfaced.
■	Explanation	User can understand why AQI is worsening/improving.

G. SIH DEMO READINESS

Check	Requirement	What to verify
■	End-to-end run	Demonstrate data → coupled model → forecast → dashboard.
■	Realistic scenario	Include an inversion/high-pollution or stubble-burning episode.
■	Real model evidence	Show actual model output, not mocked values.
■	Scientific ownership	A team member can explain the coupling and atmospheric mechanisms.
■	Performance	Forecast generation fits a practical demo workflow.
■	Fallback	Keep a recorded legitimate forecast run if live execution fails.
■	Technical ownership	Team understands every major component it claims.
■	No overclaiming	UI polish is not presented as scientific validation.

H. FINAL GO / NO-GO

Check	Requirement	What to verify
■	Can we run the coupled framework?	If not, the core PS is not satisfied.
■	Do we have the data?	Confirm meteorology, pollution and emission inputs.
■	Can we validate accuracy?	A heatmap without validation is weak evidence.

Check	Requirement	What to verify
■	Can we demonstrate coupling?	Show what changes because weather and chemistry interact.
■	Can we explain inversion effects?	This is explicitly central to the PS.
■	Can we demonstrate plume transport?	Show source → transport → Delhi NCR impact.
■	Can we produce 72 hours reliably?	Test the complete pipeline.
■	Can we defend the science?	Assign ownership for atmospheric modelling, data, backend and frontend.

FINAL ACCEPTANCE STANDARD

Minimum credible claim: A working Delhi NCR pipeline using a genuine coupled weather-chemistry framework, producing a 72-hour spatial forecast and exposing inversion and stubble-burning plume effects through a dashboard.

Strong SIH claim: The above, backed by historical validation, baseline comparison, forecast-error metrics, driver explanations and a reproducible end-to-end workflow.

Make-or-break: A conventional ML AQI predictor plus dashboard does not by itself satisfy the coupled atmospheric modelling requirement.